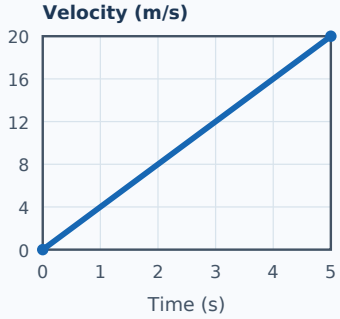
VELOCITY–TIME GRAPHS

A–H | Worked solutions

Gradient = acceleration: $a = \Delta v / \Delta t$. A straight line means constant acceleration.

Positive/negative velocity gives direction; positive/negative acceleration gives change in velocity.

Graph A

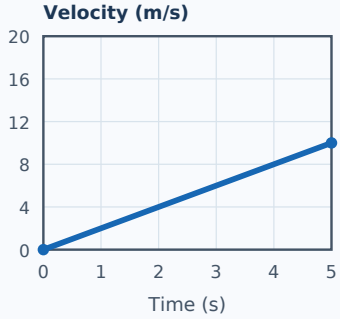


v_0 = 0 m/s
v = +20 m/s
a = (v − v_0) / t
= (+20 − (0))/5
= +4 m/s²

Starts at rest; velocity increases uniformly in the positive direction.

Acceleration constant.

Graph B

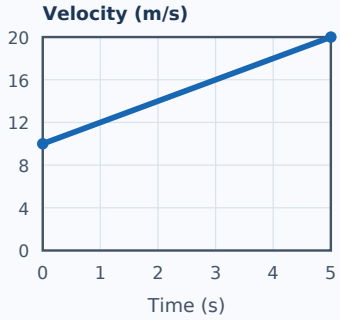


v_0 = 0 m/s
v = +10 m/s
a = (v − v_0) / t
= (+10 − (0))/5
= +2 m/s²

Starts at rest; velocity increases uniformly in the positive direction.

Acceleration constant.

Graph C

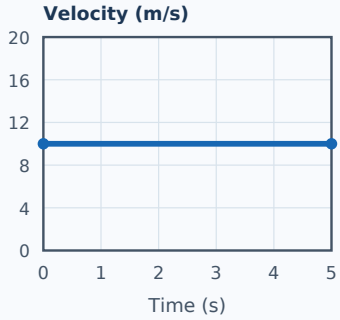


v_0 = +10 m/s
v = +20 m/s
a = (v − v_0) / t
= (+20 − (+10))/5
= +2 m/s²

Positive velocity increases uniformly; object speeds up in the + direction.

Acceleration constant.

Graph D

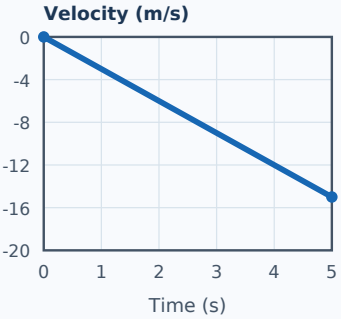


v_0 = +10 m/s
v = +10 m/s
a = (v − v_0) / t
= (+10 − (+10))/5
= 0 m/s²

Constant positive velocity of +10 m/s; no change in velocity.

Acceleration constant.

Graph E

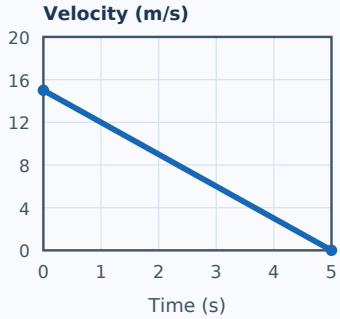


v_0 = 0 m/s
v = −15 m/s
a = (v − v_0) / t
= (−15 − (0))/5
= −3 m/s²

Starts at rest; velocity becomes more negative; speeds up in − direction.

Acceleration constant.

Graph F

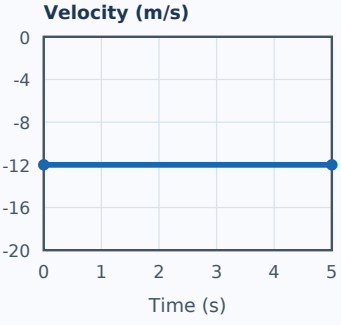


v_0 = +15 m/s
v = 0 m/s
a = (v − v_0) / t
= (0 − (+15))/5
= −3 m/s²

Positive velocity falls uniformly to zero; slows down to rest.

Acceleration constant.

Graph G

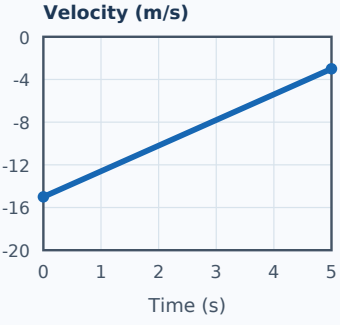


v_0 = −12 m/s
v = −12 m/s
a = (v − v_0) / t
= (−12 − (−12))/5
= 0 m/s²

Constant negative velocity of −12 m/s; speed is constant in − direction.

Acceleration constant.

Graph H



v_0 = −15 m/s
v = −3 m/s
a = (v − v_0) / t
= (−3 − (−15))/5
= +2.4 m/s²

Velocity becomes less negative; slows down while moving − direction.

Acceleration constant.